

Vedant Hariharan

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34280 Mulberry Ter, Fremont, CA 94555

Undergraduate bioengineering student interested in biocomputational engineering

EDUCATION

University of California, San Diego

Aug 2025 - Jun 2027 (est.)

University of California, Irvine

Sep 2023 - Jun 2025

Major: Bioengineering: Biotechnology, B.S., Minor: Mathematics

UC GPA: 3.944

Relevant Coursework: Advanced Python Programming, Cell and Molecular Engineering, Organic Chemistry, Fluid Dynamics, CAD & Data Analysis (Intro MATLAB), Independent Research

EXPERIENCE & PROJECTS

Haun Research Lab

Feb 2024 -

Undergraduate Researcher

- Working to create simulations of nanoparticles under fluid flow conditions in Python to replicate nanoparticle adhesion to cell membranes.
- Developing a model in Python to simulate the Brownian motion of cylindrical nanoparticles, as well as their binding/unbinding to ICAM-1 receptors, utilizing scipy, matplotlib, and pandas to interpret data.
- Utilizing UCI's High Power Community Computing Cluster (HPC3) to run high-resolution simulations.

UCSD BMES

Oct 2025 -

Bioinformatics Sublead - Bioworks Committee

- Designing bioinformatics-oriented workshops for undergraduate bioengineering students
- Workshops centered on building familiarity with bioinformatics software and databases
- Participating in a biweekly journal club

Science Gurus

Jun - Aug 2022

Cell Science Intern

- Prepared a bioinformatics presentation and research summary about the mechanisms and status of CAR-T cell therapy with a partner.
- Utilized genome databases to study expression patterns, mutations, and target traceability of CD19

Computer Aided Design Final Project

April - June 2025

Project Manager

- Designing and prototyping a novel tracheostomy tube that stabilizes against the back wall of the trachea.
 - Utilizing Solidworks and a combination of 3D-printed and store parts.
 - Culminates in a term project showcase and a CSWA certification
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LEADERSHIP

Scouts BSA, Eagle Scout (2017 -)

Troop 273, Union City, CA

Founding VP (2022 - 23)

AHS Biology Club, Fremont, CA

Peer Learning Assistant (Winter 2026)

, BENG 1, UCSD

SKILLS

Python

Solidworks

Arduino IDE/C++

Linux

UCSC Genome Browser

Team Organization

MATLAB

Problem Solving